Group B-07

Palendar

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The Problem

University students struggle to find common free times across multiple friend groups with conflicting schedules.

Key Statistics:

- 73% - Plans fail due to scheduling conflicts
- 5+ - Average minutes coordinating one hangout

Why It Matters: Social connection improves mental health, but scheduling complexity reduces opportunities for meaningful interaction. Current tools fragment communication across group chats, calendars, and manual coordination.

Data from PA2 user interviews with 8 university students

Existing Solutions

Google Calendar

- Individual scheduling, but lacks group coordination features
- ✓ Personal scheduling
- ✗ No group availability visualization

When2Meet

- One-time availability polls, no persistent groups
- ✓ Quick polls
- ✗ Single-use only

Doodle

- Event-specific polling, requires new poll each time
- ✓ Simple voting
- ✗ Repetitive setup

No solution combines persistent friend groups, automatic availability visualization, and integrated event creation for recurring coordination needs.

Competitive analysis conducted during PA1 and PA2

USER RESEARCH INSIGHTS

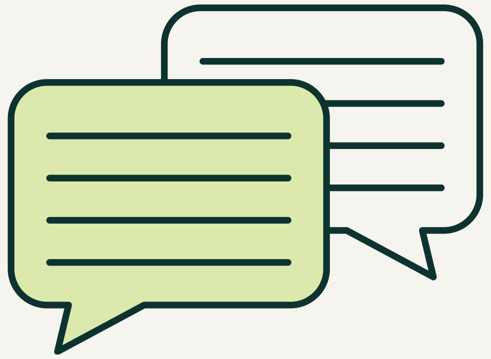
Key Findings	Must Have Features
Double-Booking Crisis "I sometimes forget I planned something with a particular group and have a tendency to double book"	✓ Personal calendar creation
Last-Minute Cancellations "Plans fall apart because of last-minute conflicts when people suddenly have gym or cricket"	✓ View group's aggregated schedule ✓ Create group events
Schedule Blindness "Hangout plans fail very frequently because people have different class times and labs"	✓ Join multiple friend groups ✓ Auto-identify common free times

Interviews & ethnography conducted in PA2 | Task descriptions using TCSD methodology

Prototype Evolution

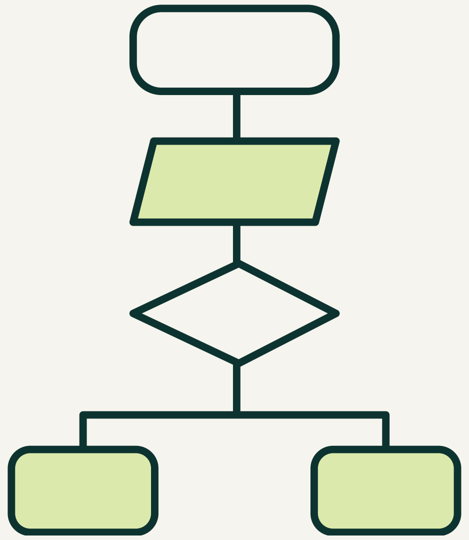
From sketches to functional interface

Brainstorming (Ideation)



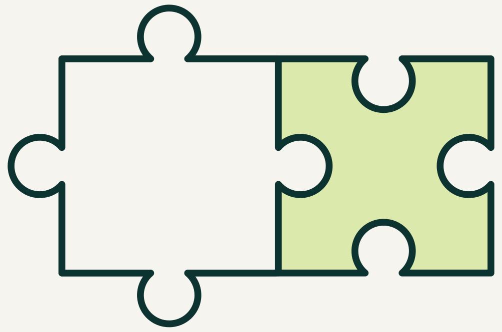
- Generated 10+ interface sketches
- Key decisions: Bottom navigation vs top navigation | Plus button vs separate create/join buttons | Color-coding system for availability

Heuristic Evaluation (Refinement)



- Identified issues before user testing
- Issues found: Insufficient system status feedback | Color-coding confusion | Interface clutter | Recognition vs recall problems

Interactive Prototype (Validation)



- Web prototype ready for testing
- Improvements: Green indicators for availability | Minimalist design | Persistent navigation | Clear confirmation messages

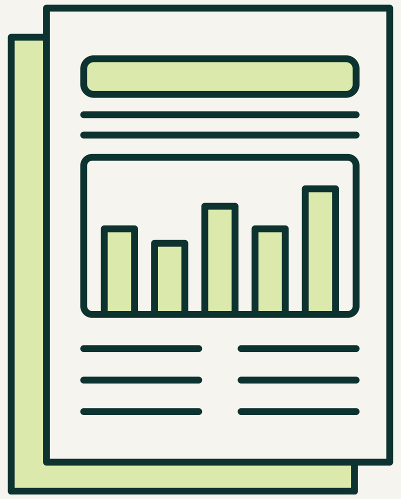
Iterative design process following HCI best practices

Key Design Improvements

From PA3 heuristic evaluation to PA4 prototype

Visibility of System Status

BEFORE: No feedback on actions
AFTER: Explicit confirmations - Green icons, success messages



Color-Coding Clarity

BEFORE: Inconsistent semantics
AFTER: Consistent green = Universal color semantics throughout app

Interface Simplification

BEFORE: Many buttons and features visible
AFTER: Minimalist approach - Focus on essential actions only

User Evaluation Results

4 participants | 100% task completion

Positive Feedback:	"The buttons were where I would expect them to be... as straightforward as I can imagine" - Participants 1 & 2
	“She was able to do all tasks without any intervention, a big W in my opinion" - Facilitator observation
Areas for Improvement:	Navigation Discoverability - Bottom placement not immediately obvious
	Logout icon confused with profile/home
	Want granular info at-a-glance
Key Learning:	Task completion ≠ optimal UX. Even "straightforward" interfaces benefit from refinement.

PA5 usability testing | 8-phase protocol from PA4 evaluation plan

Lessons Learned

What Worked Well:

✓ Task-centered design methodology translated user needs into functional solutions
✓ Iterative prototyping caught issues early (heuristic evaluation before user testing)
✓ Think-aloud protocol revealed user reasoning and mental models
✓ Structured evaluation plan maintained consistency across 4 sessions

What Surprised Us:

- Navigation expectations vary by platform - Bottom navigation works on mobile, but desktop users looked at the top first
- Icon affordances matter - Person icon = profile, not logout. Convention violations cause confusion even when design is "intuitive"

What We'd Improve With More Time:

- Implement availability heat map (color intensity for granular info)
- Test with larger, more diverse participant pool
- Build Google Calendar integration
- Conduct longitudinal evaluation (sustained real-world use)
- Replace logout icon with conventional signaling
- Add nightmode for accessibility

User feedback should directly drive design iteration — specific, actionable insights are the reward of user-centered methodology

Future Work

Priority 1: Navigation & Icon Refinement

- Graduated color-coding with numerical indicators
- At-a-glance availability assessment

Priority 2 : Calendar Integration & Aesthetics

- Enhance bottom navigation discoverability
- Replace logout icon with conventional signaling

Long-Term Research

- User adoption patterns
- Feature engagement drivers
- Impact on social interaction

Priorities informed by PA5 user feedback

Contributions & Impact

For HCI Research:

- Platform-specific design matters
- Icon conventions are critical

For Students & Industry:

- Persistent groups solve real problems
- Network effects drive adoption

For Society:

- Reduces scheduling friction
- Supports social connection and mental health

Conclusion: User-centered design produces interfaces that genuinely address human needs.